

Part A – Conceptual Understanding (Theory)

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1. What are Supervised Learning Algorithms?

Supervised Learning is a type of machine learning in which a model learns from labeled data. The dataset contains both input features (independent variables) and the correct output (dependent variable). The model learns the relationship between the inputs and outputs and uses this knowledge to make predictions on new data.

Examples:

- Predicting house prices
 - Email spam detection
 - Loan approval prediction
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2. Difference Between Regression and Classification Algorithms

Regression

Predicts continuous numerical values.

Output is numeric.

Used for price prediction, sales forecasting.

Example: House Price Prediction.

Classification

Predicts categorical values or classes.

Output is a category or label.

Used for spam detection, disease prediction.

Example: Email Spam Detection.

3. Explain Simple Linear Regression

Simple Linear Regression is a supervised learning algorithm that models the relationship between one independent variable (X) and one dependent variable (Y). It predicts the target value by fitting the best straight line through the data.

Equation

$$Y = mX + c$$

Where:

- **Y** = Predicted value
- **X** = Independent variable
- **m** = Slope of the line
- **c** = Intercept

Example: Predicting house price using house area.

4. Assumptions of Linear Regression

Linear Regression is based on the following assumptions:

1. **Linearity:** There should be a linear relationship between independent and dependent variables.
 2. **Independence:** Observations should be independent of each other.
 3. **Homoscedasticity:** The variance of residuals should remain constant.
 4. **Normality of Residuals:** Residuals should follow a normal distribution.
 5. **No Multicollinearity:** Independent variables should not be highly correlated with each other (important in multiple regression).
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5. What is Bias-Variance Trade-off?

The Bias-Variance Trade-off is the balance between underfitting and overfitting.

- **High Bias:** The model is too simple and underfits the data.
- **High Variance:** The model is too complex and overfits the training data.

A good machine learning model maintains an appropriate balance between bias and variance to achieve good performance on unseen data.

6. Explain Overfitting and Underfitting with Examples

Overfitting

Overfitting occurs when a model learns the training data too well, including noise and small fluctuations. It performs very well on training data but poorly on new data.

Example: A polynomial regression model with a very high degree that perfectly fits the training data but gives poor predictions on unseen houses.

Underfitting

Underfitting occurs when a model is too simple to capture the underlying relationship in the data. It performs poorly on both training and testing data.

Example: Using a straight line to model a complex non-linear relationship between house area and price.